
MATH 562: STATISTICAL INFERENCE

Homework 2

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1 Preliminaries

The aim for this section is to introduce some statistical notions that will be used in order to solve the exercises in this homework. We begin exploring the concept of a *statistical pivot*. A statistical pivot (or *pivotal quantity*) is a function of both the sample data and one or more unknown parameters whose **probability distribution does not depend on those parameters**.

Definition 1.1 (Statistical Pivot). Formally, if $X = (X_1, \dots, X_n)$ is a random sample from a distribution depending on a parameter θ , a statistic $T(X, \theta)$ is called a *pivot* if the distribution of T is the same for all possible values of θ .

In simpler terms, a pivot is a clever combination of our data X and the unknown parameters θ that “cancels out” the dependence on those parameters, leaving something whose behavior is fully known. As we’ll explore throughout the worked exercises, pivotal quantities are extremely useful in statistical inference because they allow us to construct **confidence intervals** and perform **hypothesis tests** without knowing the true value of the underlying parameter.

Remark 1.2. By using a pivot, we can make probability statements that remain valid for any possible value of θ .

Intuition: We may think of a pivot as a “universal yardstick.” It transforms the random variability in your sample into a standardized form that behaves predictably, regardless of the true value of the parameter. This universality is what makes pivots so powerful in practice.

We now explore the concept of *equivariant estimators*. Equivariant estimators are statistical estimators that maintain a consistent relationship with the data under certain transformations.

Definition 1.3. A parameter estimator $g(y)$ is said to be **equivariant** if it satisfies:

$$g(a + by) = g(a + by_1, \dots, a + by_n) = a + g(by_1, \dots, by_n) = a + bg(y)$$

for all $a \in k$ and $b > 0$.

It is useful to recall the following remark regarding the equivariance properties of estimators in a location-scale model.

Remark 1.4 (Equivariance properties of estimators). For estimators in a location-scale model

$$Y_i = \eta + \tau V_i, \quad i = 1, \dots, n,$$

it is natural to require that they behave consistently under changes of location and scale. A **location estimator** $m(\cdot)$ is said to be *location-scale equivariant* if

$$m(\eta + \tau v_1, \dots, \eta + \tau v_n) = \eta + \tau m(v_1, \dots, v_n),$$

that is, translating and rescaling the data simply translates and rescales the estimator in the same way.

Similarly, a **scale estimator** $s(\cdot)$ is said to be *location invariant* and *scale equivariant* if

$$s(\eta + \tau v_1, \dots, \eta + \tau v_n) = |\tau| s(v_1, \dots, v_n).$$

Since scale parameters are typically assumed to be positive in practice, we often write this as

$$s(\eta + \tau v_1, \dots, \eta + \tau v_n) = \tau s(v_1, \dots, v_n).$$

These properties ensure that both estimators respond to shifts and rescalings of the data in a coherent way, preserving the structure of the underlying model.

2 Exercise

2.1 Defining the problem

A *location-scale* model is of the form $y_0 = \eta + \tau v_0$ where η is the location parameter (a scalar i.e. $\eta \in k$ for an arbitrary field k), $\tau > 0$ is the scale parameter and v_0 is a random variable with known distribution. Given a *random sample* $y = (y_1, \dots, y_n)$ with the y_i 's being independent, identically distributed (*i.i.d*) random variables from the population y_0 , we may produce estimators $m(y), s(y)$ for the parameters η, τ respectively.

- Show that $q_1(y, \eta) = \frac{m(y) - \eta}{s(y)}$ and $q_2(y, \tau) = \frac{s(y)}{\tau}$ are pivots. Moreover, explain how could we construct confidence intervals for the parameters η, τ using these pivots.
- Deduce that pivots can be formed by taking:
 - $m_1(y) = \bar{y}$ and $s_1(y) = \sqrt{\sum (y_i - \bar{y})^2}$.
 - $m_2(y) = \text{median}(y)$ and $s_2(y) = \mathbf{IQR}(y)$ i.e. the interquartile range.
- Provide a pivot that could be used to construct a prediction interval for y_0 based on $y_1, \dots, y_n$.

2.2 Solution

In order to verify that q_1, q_2 are pivots, we need to show that their distributions do not depend on the parameters η, τ . In order to do so, we will use the fact that m, s are equivariant estimators of location and scale respectively. We write:

$$\begin{aligned} q_1(y, \eta) &= \frac{m(y) - \eta}{s(y)} = \frac{m(y_1, \dots, y_n) - \eta}{s(y_1, \dots, y_n)} = \frac{m(\eta + \tau v_1, \dots, \eta + \tau v_n) - \eta}{s(\eta + \tau v_1, \dots, \eta + \tau v_n)} \\ &\Rightarrow q_1(y, \eta) = \frac{\eta + \tau m(v) - \eta}{|\tau|s(v)} = \frac{\tau m(v)}{\tau s(v)} = \frac{m(v)}{s(v)} \end{aligned}$$

hence the distribution of q_1 does not depend on η, τ and is in fact the distribution of $\frac{m(v)}{s(v)}$. Similarly, we have:

$$\begin{aligned} q_2(y, \tau) &= \frac{s(y)}{\tau} = \frac{s(y_1, \dots, y_n)}{\tau} = \frac{s(\eta + \tau v_1, \dots, \eta + \tau v_n)}{\tau} \\ &\Rightarrow q_2(y, \tau) = \frac{|\tau|s(v)}{\tau} = \frac{\tau s(v)}{\tau} = s(v) \end{aligned}$$

hence the distribution of q_2 does not depend on η, τ and is in fact the distribution of $s(v)$.